

ARNAV SHARMA

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EDUCATION

Bachelor of Technology — Computer Science

Aug 2023 – May 2027

Manav Rachna International Institute of Research and Studies, Haryana

TECHNICAL SKILLS

Languages: Java, Python, SQL

Frameworks & Libraries: React, Node.js, Express, PyTorch, Hugging Face Transformers & Datasets, Spring Boot, Three.js, Plotly.js, Math.js, Tailwind CSS

AI / ML: LLM Fine-tuning (QLoRA, DPO, PPO), RLHF, Reward Modeling, Supervised Fine-tuning

Cloud & DevOps: AWS (Bedrock, DynamoDB, S3, Amplify), Firebase (Auth, Hosting), Git, GitHub, Vercel, Render

Tools: npm, Postman, Kaggle, Vite

Core CS: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS

PROJECTS

Mini-InstructGPT — LLM Alignment Pipeline

2026

- Built an end-to-end LLM alignment pipeline in Python/PyTorch covering Supervised Fine-Tuning (SFT), reward modeling, and PPO-based RLHF stages.
- Diagnosed and resolved reward-hacking behavior in GPT-2 PPO runs by iterating on training and evaluation scripts.
- Scaled experiments to Qwen2.5-7B-Instruct using QLoRA + DPO and benchmarked outputs on a fixed 100-prompt evaluation set.
- Produced reproducible evaluation artifacts (JSONL/CSV) enabling side-by-side model comparison across alignment methods.

Interactive Tech Tutor — AI-Powered Learning Platform

2025

- Designed and shipped a full-stack AI tutoring platform with a React + TypeScript (Vite) frontend and Node.js + Express backend generating step-by-step visual simulations for DSA, ML, and OS concepts.
- Integrated AWS Bedrock, DynamoDB, S3, and Amplify for AI simulation generation, data persistence, and deployment with JWT-based authentication and session/history handling.
- Implemented interactive visual simulations using HTML5 Canvas and Three.js with voice narration and voice-command controls via the Web Speech API.

MindSpace AI — Wellness Prototype

2025

- Developed a Next.js + React + TypeScript wellness application featuring role-play chat, goal planner, and guided relaxation modules.
- Integrated Firebase Authentication and Hosting with privacy-first client-side data flows using React Context and local storage.
- Designed a responsive, animated UI using Tailwind CSS and Framer Motion for smooth, engaging user interactions.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle
- Building LLM Applications With Prompt Engineering — NVIDIA
- Prompt Design in Vertex AI Skill Badge — Google
- Core Java - Internshala

ADDITIONAL INFORMATION

Languages: English (Fluent), Hindi (Native), Spanish (Conversational)

Interests: AI/ML Research